



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

EEE 127: Electronic Devices and Circuits

Batch: 24th

Semester: Fall 2023

Date: 08/12/2023

Mid-Term Examination

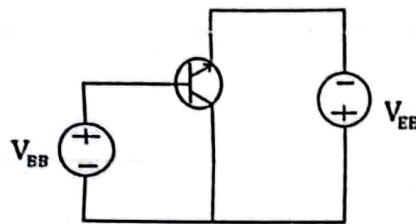
Time: 1 Hour 30 minutes

Total Marks: 30

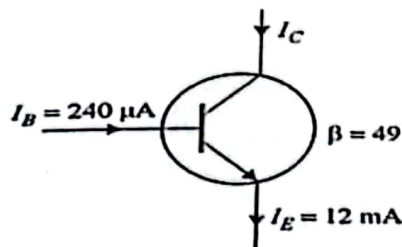
Answer any five questions of the following.

5 × 6 = 30

1. (a) Write the relationship equations between α and β . CLO2 2
(b) Explain band theory with the necessary diagrams. CLO1 4
2. (a) Suppose you have a piece of impure semiconductor, describe the procedure to produce the first electronic device from it. CLO1 3
(b) To operate the p-n junction, we have some limitations, mention them. CLO1 3
3. (a) How Transistors are built? Mention the Terminals name. CLO2 3
(b) What is Potential Barrier? Why does it exist in a junction? CLO2 3



- (a) What type of transistor configuration does the circuit represent? Show the current direction of the circuit. CLO2 3
(b) Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu A$. CLO3 3



- (a) Write two properties of Semiconductor. CLO1 2
(b) Find the α rating of the transistor shown in the figure. Hence determine the value of I_C using both α and β rating of the transistor. CLO3 4

- / (a) Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$. CLO3 2
- (b) Describe three types of transistor configuration with appropriate figures. CLO2 4
- / (a) In a common base connection, the current amplification factor is 0.9. If the emitter current is 1mA, determine the value of base current. CLO3 2
- (b) Explain the I-V Curve of p-n junction for both Forward and Reverse Bias. CLO2 4

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Midterm Examination (Fall'23)

PHY 121 (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) from the following sets of questions

- 1 (a) State Coulomb's Law. Prove it using Gauss' Law. 5 CLO1
- (b) Four equal point charges, $+3\text{mC}$, are placed at the four corners of a square that is 40 cm on a side. Find the force on any one of the charges. 5 CLO4
- 2 (a) What is equipotential surface. Explain with necessary figures and equation 3 CLO1
- (b) Determine the electric field due to uniformly charged wire by using both Coulomb's law and Gauss' Law. Explain from that the effectiveness of using Gauss' Law. 7 CLO1
- 3 (a) 3 charges are arranged as in the Fig.3.(a). What is their mutual potential energy? Assume that $q = 1 \times 10^{-7}$ coulomb and $a=10$ cm. 5 CLO4

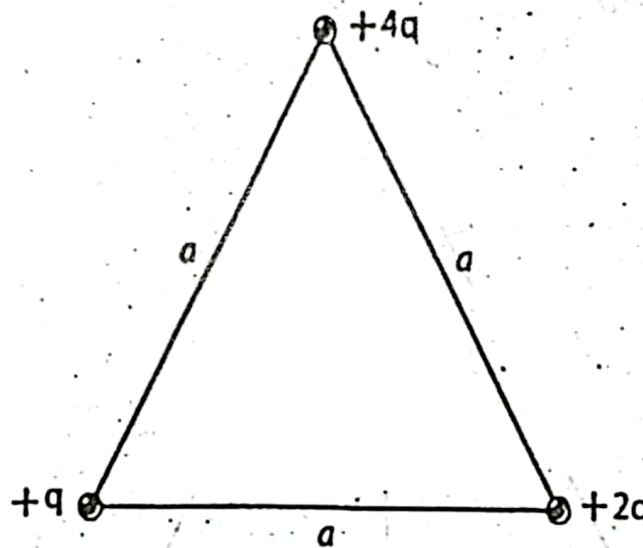


Fig.3. (a)

- (b) Derive the equation for Electric Potential Energy. 5 CLO1

- 4 (a) What is an electrical dipole? Prove that $U = -PE (\cos \theta - \cos \theta_0)$ (where each symbol used for usual meaning) 7 CLO1
- (b) An electric dipole consists of two opposite charges of magnitude 2 micro coulomb, separated by a distance $l = 1$ cm. It is placed in an external electric field of 22×10^5 N/C. What maximum torque does the field exert on the dipole? 3 CLO4



Faculty of Science, Engineering and Technology
Department of Electrical and Electronics Engineering
CSE 121: Structured Programming Language

Batch: 24th

Semester: Fall 2023

Date: 15/12/2023

Mid-Term Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions of the followings.

5 × 6 = 30

1. (a) Write some features of Structured Programming Language. 2
(b) Write a program in C to perform addition, subtraction, multiplication and division of two numbers where number inputs are given by the user. 4
2. (a) Explain the header files in C in brief with example. 2
(b) What is data type in C? Explain the basic data types of C programming. 4
3. (a) Briefly describe prefix and postfix operator in C with example. 2
(b) Write a program in C that takes N number of inputs from users and calculate the sum of odd and even numbers of N numbers without using an array. 4

Sample input/output:

Enter the value of size

4

Enter 4 numbers

2

4

3

5

Sum of all odd numbers = 8

Sum of all even numbers = 6

4. (a) Write the basic structure for 'while' and 'do...while' loop. 2
(b) Write a program in C for the following pattern: 4

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

5. (a) What are two dimensional arrays? Write a program in C to input and output a 2×3 matrix. 3
(b) Write a program in C for the following series: 3

$$2^2 + 4^2 + 6^2 + \dots + n^2$$

6. (a) Define format specifier with examples. 2
(b) Write a program in C to find the maximum and minimum value of an array. 4

7. (a) What is call by value and call by reference in C? 2
- (b) Write a C function for a given number is Palindrome or not. 4

Hamdard University Bangladesh

Dept. of EEE, CSE and Mathematics
Lab Final Examination (Fall'23)

Course code and Title:
Time: 1-hour

02
Marks: 40

Answer all the Questions from the following sets of questions

1. Write down the name of the experiments you have performed in your physics lab. 03
2. Suppose, you have a sphere and a cube. Now, for measuring the length of the cube, you found the main scale reading of 5 cm and vernier scale reading of 0.3 cm with a negative error of 0.5cm. The diameter of the sphere is the last two digits of your Student ID (unit in mm). Suddenly, you insert the sphere into the cube. Find the volume of the void portion after inserting the sphere into the cube. 05
3. Look the following figure. Now, calculate the Time period, frequency, extension while the spring is in equilibrium, equation of the displacement at $t=0.05s$. Assume that, the surface is smooth and the air friction is negligible. [Don't Panic!!!!... :P] 08

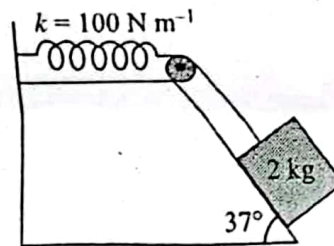
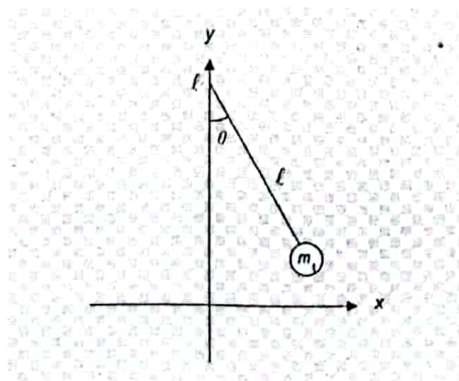


Fig.3

4. What type of problems you faced in your lab? 02
5. The mass, length and angle of the following Simple pendulum are 0.3 kg , 10 cm and 30 degree. Find the tension force of the rope of the pendulum. Assume that, the pendulum is on a running bus with an acceleration of 2 ms^{-2} . 05



6. Fill the appropriate answer in the blanks given below: 03
- The value of g in ideal case is cms^{-2}
 - The number of vernier scale is 100. The VC is.....
 - If we connect three identical springs in series with a constant of K the, the equivalent spring constant is.....
7. Mark the following statements as "T" for True and "F" for False. 03
- T^2 vs L graph of a simple pendulum is parabolic.
 - m vs k graph of a simple pendulum is straight line.
 - Negative error of a slide calipers is always subtractive.
8. Write down the correct answer from the Multiple-Choice Questions given below: 04
- Vernier calipers helps in measuring-

(i) External diameter	•(ii) Internal diameter
(iii) Thickness	(iv) All of them
 - Spring constant is dependent on-

(v) mass	•(vi) the value of g
(vii) material of spring	(viii) All of them
 - If a spring is vibrated in a downward lift, then, the value of g will be-

(ix) increased	•(x) decreased
(xi) same as before	(xii) none of the above
 - What do you think about me? -

(xiii) Bad	(xiv) Very bad
(xv) Worse	(xvi) Worst
9. Elucidate briefly on your experience of this laboratory. 03

PAHIMHND UNIVERSITY DHANBADH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
BSC in Computer Science & Engineering
ENG 127: English Skills
Midterm Examination
Fall 2023

24th Batch
22 December 2023

2nd Semester
Time: 1 hour 30 minutes

Total Marks: 30

Section A

CLOs Level Marks

Read the following passage and answer questions 1 & 2:

Countries of the world rely heavily on petroleum, coal and natural gas for their energy sources. There are two major types of energy sources: renewable and non-renewable. Hydrocarbon or fossil fuels are non-renewable sources of energy. Reliance on them poses real big problems. First, fossil fuels such as oil, coal, gas, etc. are finite energy resources and the world eventually will run out of them. Secondly, they will become too expensive in the coming decades and too damaging for the environment. Thirdly, fossil fuels have direct polluting impacts on earth's environment causing global warming. In contrast, renewable energy sources such as, wind and solar energy are constantly and naturally replenished and never run out.

Most renewable energy comes either directly or indirectly from the sun. Sunlight or *solar energy* can be used for heating and lighting homes, for generating electricity and for other commercial and industrial uses.

The sun's heat drives the wind and this wind energy can be captured with wind turbines to produce electricity. Then the wind and the sun's heat cause water to evaporate. When the water vapor turns into rain or snow and flows downhill into rivers or streams, its energy can be captured as *hydroelectric energy*.

Along with the rain and snow, sunlight causes plants to grow. Plants produce biomass which again can be turned into fuels such as firewood, alcohol, etc. that are called *bioenergy*.

Scientists have identified *Hydrogen* as another form of renewable energy source. It is the most abundant element in nature. But it does not exist separately as a gas. It is always combined with other elements, such as with oxygen to make water. Hydrogen, separated from another element, can be burned as a fuel to produce electricity.

Our Earth's interior contains molten lava which gives off extreme heat. This heat inside the Earth produces steam and hot water which can be used as *geothermal energy* to produce electricity for heating homes, etc.

Ocean energy comes from several sources. Ocean's force of tide and wave can be used to produce energy. The surface of the ocean gets more heat from the sun than the ocean depths. This temperature difference can be used as energy source too.

1. Choose the correct answer from the following alternatives:

- (a) Solar energy is the source of ____.
- (i) non-renewable energy
 - (ii) renewable energy
 - (iii) hydroelectric energy
 - (iv) geo-energy
- (b) Consumption of fossil fuels ____.
- (i) is favourable for our happy life in the world.
 - (ii) is causing irreparable damage to our environment.
 - (iii) does not create any problem for us.
 - (iv) is not likely to lead us towards a crisis.
- (c) What is the message of the text?
- (i) to show the destruction of non-renewable energy
 - (ii) to put up eco-friendly character of natural oil and gas
 - (iii) to present positive sides of fossil fuels
 - (iv) to relate awareness about using renewable energy sources
- (d) The sun is responsible ____.
- (i) for the ruin of the world
 - (ii) to cause evaporation of sea water and than rain
 - (iii) for our environment pollution
 - (iv) for our many difficulties
- (e) Wind and sun's heat jointly result in ____.
- (i) distillation
 - (ii) replenishment
 - (iii) evaporation
 - (iv) decantation
- (f) Constantly means ____.
- (i) regularly
 - (ii) normally
 - (iii) irregularly
 - (iv) occasionally
- (g) The word resources mean ____.
- (i) Sources
 - (ii) sourceable
 - (iii) profit
 - (iv) wealth
- (h) Hydrogen is the most abundant in nature which is the part of speech underlined the word.
- (i) noun
 - (ii) adjective
 - (iii) adverb
 - (iv) verb
- (i) What is the opposite word of exist?
- (i) dwell
 - (ii) die
 - (iii) depart
 - (iv) finish
- (j) Which parts of speech used both noun and verb?
- (i) gas
 - (ii) fuel
 - (iii) water
 - (iv) oil

2. Answer the following questions from the above passage:

CLO1,2 Understand 6x1=6

- a) Why is solar energy so important and why should it be used?
- b) What is the impact of non-renewable energy?
- c) What are the differences between renewable and non-renewable energy sources?
- d) What is the geothermal energy and how can it be used?
- e) Why do you think that reliance on non-renewable sources of energy will cause great problem?
- f) What can be burned as a fuel to produce electricity?

Section B

3. Answer any two of the following:

CLO1 Understand 7x2=14

- a) Briefly discuss the types of reading with examples.
- b) Elaborate the strategies of listening.
- c) What are the purposes of speaking? Write down its characteristics.



Midterm Examination (Fall 2023)

Faculty of Science, Engineering, and Technology	Department of Computer Science and Engineering(CSE)		
Course: MAT129 Linear Algebra and Complex Variable	Batch: 23 th	Semester: 2nd	Course Teacher: Md. Jamiul Islam

Time: 90 Minutes

Total Marks: 30

Answer any 3 (Three) from the following sets of questions

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

Sl. No.	Questions	CLOs	Marks
1.	(a) Define homogeneous system of linear equation with example.	CLO1	01
	(b) Solve the following homogeneous system of linear equations by using Gauss-Jordan elimination method.	CLO2	06
	$\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \end{aligned}$		
	(c) Solve the following system of linear equations by row echelon form:	CLO2	03
	$\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$		
2.	(a) What is inverse and invertible matrix?	CLO1	01
	(b) Find the solution of linear system by using inverse matrix, the following equations:	CLO2	05
	$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17 \end{aligned}$		
	(c) Using row operation to find A^{-1} by the following matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$	CLO2	04
3.	(a) Define linear functional and dual space.	CLO1	02
	(b) If $B = \{(-1, 3), (2, -5)\}$ be a basis and vector space of R^2 , then find the dual basis of B .	CLO2	04
	(c) If $x = (x_1, x_2)$ and $y = (y_1, y_2)$ be in R^2 , then prove that $f(x, y) = 2x_1y_2 - 3x_2y_1$ is a bilinear form on R^2	CLO2	04
4.	(a) What is Eigenvalue and Eigenvector?	CLO1	02
	(b) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$	CLO2	02
	(c) Find the Eigenvalue and Eigenvector of the following matrix :	CLO2	06
	$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$		



Faculty of Science, Engineering and Technology
Department of Electrical and Electronics Engineering
CSE 121: Structured Programming Language

Batch: 24th

Semester: Fall 2023

Date: 16/03/2024

Time: 2 Hours

Final Examination

Total Marks: 40

Answer five questions of the followings.

5 × 8 = 40

1. (a) Define a function. Write about the advantages of using functions in C programming. 3
 (b) What is a recursive function in C programming? Write a program in C to find the factorial of a given number using recursion. 5

2. Find the error in each of the following program segments and explain how the error can be corrected : 4×2=8

i. `int g( void )`
`{`
`printf( "Inside function g\n" );`
`int h( void )`
`{`
`printf( "Inside function h\n" );`
`}`
`}`

ii. `int sum( int x, int y )`
`{`
`int result;`
`result = x + y;`
`}`

iii. `int sum( int n )`
`{`
`if ( n == 0 ) {`
`return 0;`
`}`
`else {`
`n + sum( n - 1 );`
`}`
`}`

iv. `void f( float a );`
`{`
`float a;`
`printf( "%f", a );`
`}`

3. (a) Why are the *gets()* and *puts()* functions used in C programs? 2
 (b) Write a C program that can take a sentence as input and show the words in separate line. 6

Sample Input: We are CSE 24

Sample Output: We

are

CSE

24

and count number of vowels in the given sentence.

4. (a) Write about built-in functions and user-defined functions in C programming. 2

- (b) Consider the following two matrix A and B:

$$A = \begin{bmatrix} 5 & 6 & 7 \\ 3 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Write a program in C to perform the matrix addition between A and B.

5. (a) What are the differences between structures and unions in C? 3
(b) Write a C structure named **student** which include **roll**, **name**, **cgpa** and **address** 5
and then take the input for 5 students and show the info of 5 students.
6. (a) What are the basic operations of file in C? Write about the basic file modes in C. 4
(b) Write a C program to read name and marks of n number of students and store 4
them in a file. If the file previously exists, add the information to the file.
7. (a) Define pointer in C. 2
(b) What does the symbols '*' and '&' mean and when they are placed in front of an 2
identifier?
(c) Write a C program to addition and subtraction of two numbers using pointer. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

Batch: 24th

Semester: Fall 2023

Date: 09/03/2024

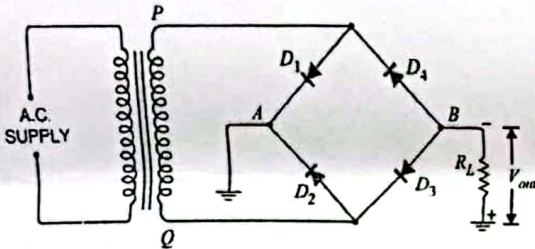
Final Examination

Time: 2 Hour

Total Marks: 40

Answer any five questions:

5 × 8 = 40

1. (a) Why do we use Zener Diode? CLO3 2
(b) Describe three types of transistor configuration with figures. CLO2 6
2. (a) Explain half-wave and full-wave bridge rectifiers with the necessary diagrams and signals. CLO4 4
(b) Derive the equations for Average voltage and RMS voltage in a Half-Wave Rectifier. CLO4 4
3. Consider the following figure:
- 
- (a) Calculate PIV for the center-tapped rectifier. CLO4 3
(b) This is a step-down transformer with a 15:1 ratio. The input voltage is 220V and the frequency is 60Hz. Now calculate the DC output Voltage. CLO3 5
4. (a) Write the Differences of JFETs compared to BJTs. CLO4 4
(b) State the advantages and disadvantages of Transistor. CLO2 4
5. (a) What is Pinch Off Voltage? CLO4 3
(b) Calculate the change in Zener voltage if Zener voltage is 15v at room temperature and new Temperature is 100⁰ C. Temperature co-efficient (T_c) is 0.072%. CLO3 5
6. (a) Draw the Construction Circuit for both DE-MOSFET and E-MOSFET. CLO4 4
(b) For the circuit given in question number 3, find Rectification Efficiency and Output Frequency. CLO3 4

7. (a) Write four Applications of FET.
(b) Draw and briefly discuss the Construction Circuit and Output Characteristic curve for n-channel JFET.

CLO4 3
CLO4 5



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering (CSE)

B.Sc. in Computer Science and Engineering

Faculty of Science, Engineering and Technology

Course Code: MAT129

Course Title: Linear Algebra and Complex Variable

Final Examination

Semester: Fall 2023

Date: 3/23/2024

Total Time: 2.0 hours

Total Marks: 40

Answer any four (04) of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) What is complex number? [01]
 (b) Define modulus and argument of a complex number and find the modulus and argument of the following complex numbers: [06]
 (i) $z = \left(\frac{1+i}{1-i}\right)^2$,
 (ii) $z = \frac{-2}{1+i\sqrt{3}}$.
 (c) For complex numbers $z_1, z_2, z_3, \dots, z_n$, then show that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\{|z_1|^2 + |z_2|^2\}$. [03]
2. (a) What is complex function? [02]
 (b) Define an analytic function and determine whether $f(z) = (4x^2 - 6xy + 3) + i(8xy - 3y^2 + 5x)$ is a analytic or not. [04]
 (c) Prove that Cauchy-Riemann equations are satisfied the following functions $f(z) = e^{z^2}$. [04]
3. (a) Define harmonic function with example. [01]
 (b) Solve the following equations: [04]
 (i) $x^6 - 3x^3 + 3 = 0$.
 (ii) $x^7 + x^4 + x^3 + 1 = 0$.
 (c) Define harmonic conjugate function and show that the function $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function. Also find harmonic conjugate of u . [05]
4. (a) What is complex integration? [01]
 (b) State and prove Cauch's integral formula. [05]
 (c) Find $\oint_c \frac{\sin^6 z}{(z - \frac{\pi}{6})^3}$ where c is the circle $|z| = 1$. [04]
5. (a) Define singular point, isolated singular point and pole. [02]
 (b) Locate and name the singularities in the finite z plane of the function $f(z) = \frac{z}{(z^2 + 4)^2}$. [03]
 (c) If we choose the principal branch of $\sin^{-1} z$ to be that one for which $\sin^{-1} 0 = 0$, then prove that $\sin^{-1} z = \frac{1}{i} \ln(iz + \sqrt{1 - z^2})$. [05]
6. (a) State Taylor and Laurent series. [02]
 (b) Expand $f(z) = \ln(1 + z)$ in a Taylor series about $z = 0$. [04]
 (c) Determine the Laurent series for $f(z) = \frac{1}{z+5}$ that are valid in the region
 (i) $|z| < 5$
 (ii) $|z| > 5$

4
4

$$x^3 \quad nx^{n-1} \quad 3x^{3-1}$$

$$8y - 6x \quad z^2 + 2 \cdot 1 \cdot i + i^2 \quad i^2 - 2 \cdot 1 \cdot i + i^2 \quad 1 + 2i - 1 \quad 1 - 2i - 1$$

$$(x+iy)^2 = x^2 + 2 \cdot x \cdot iy + (iy)^2 = x^2 + 2ixy - y^2 \quad \frac{(2i)(2i)}{(-2i)(2i)} \quad \frac{(2i)(2i)}{(-2i)(2i)}$$



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology
BSC in Computer Science & Engineering
ENG 127: English Skills
Final Examination
Fall 2023

24th Batch
20 March 2024

2nd Semester
Time: 2 Hours

Total Marks: 40

Questions

1. Answer any one of the following:

- Discuss the techniques of paragraph development.
- What is a book review? Write down the steps for a book review.

CLOs	Level	Marks
CLO3	Understand	08

2. Rewrite the following passage into its correct order:

The ant moved towards the leaf and climbed up onto it. Seeing that the ant was in trouble, the dove quickly plucked a leaf and dropped it into the water near the struggling ant. She could have drowned if a dove up a nearby tree had not seen her. One hot day, an ant was searching for some water, after walking around for some time, she came to a spring. While making her way up, she slipped and fell into the water. To reach the spring, she had to climb up a blade of grass. The leaf drifted to dry ground, and the ant jumped out, and she was safe at last.

CLO2	Understand	07
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3. Write a summary of the following passage within 50 words and give a title of the passage:

Bangladesh is an agricultural country. Most of the people live in villages and are farmers. They grow cereals, pulses, vegetables and fruits. The farmers lead a tough life. They get up early in the morning and go to the fields. They stay and work on the farm late till evening. The farmers usually live in small huts. Though, they work hard they remain poor. Farmers eat simple food; wear simple clothes and rear animals like cows, buffaloes and oxen. Without them there would be no crops for us to eat. They play an important role in the growth and economy of a country.

CLO2	Understand	07
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4. Write a review of a book that you have recently read (in approximately 250-300 words).

CLO4	Apply	10
CLO5		

5. Write a paragraph on 'A Smartphone' within 150-200 words.

CLO4	Apply	08
CLO5		

Grenna

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Final Examination (Fall'23)

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Marks: 40

Time: 2 hours

Answer any 4 (Four) from the following sets of questions

- 1 (a) Define magnetic field. Determine the force on a single charged carbon ion moving with a speed of 3×10^5 m/s at right angles to a magnetic field of 7500G? What is the centripetal acceleration? 6 CLO2
- (b) Define Hall Effect with necessary equation and figures. 4 CLO2
- 2 (a) How Maxwell speculated that light is an electromagnetic wave produced by the acceleration of charges? 4 CLO2
- (b) Two long and fixed parallel wires, A and B are 10 cm apart in air and carry currents of 40 and 20 A, respectively opposite directions. Determine the resultant flux density (a) on a line midway between the wires and parallel to them, and (b) on a line 8 cm from wire A and 18 cm from wire B. 6 CLO2
- 3 (a) Define Lenz's law. Discuss it. 3 CLO2
- (b) A 130-turn close-packed coil of diameter 2.1 cm is placed at the centre of a long solenoid which has 220 turns/cm and carries a current $i = 1.5$ A; its diameter is 3.2 cm. The current in the solenoid is increased from zero to 1.5A at a steady rate over a period of 0.16s. What is the absolute value of the induced emf that appears in the central coil while the current in the solenoid is being changed? 7 CLO2
- 4 (a) Define Mutual inductance. Construct the equation for it. 5 CLO2
- (b) A flux of 0.5 mWb is produced by a coil of 900 turns wound on a ring with a current of 3A in it. Calculate (i) the inductance of the coil, (ii) the emf induced in the coil when a current of 5A is switched off, assuming the current to fall to zero in 1 millisecond and (iii) the mutual inductance between the coils, if a second coil of 600 turns is uniformly wound over the first coil. 5 CLO2
- 5 (a) Define time dilation. Prove that a clock in moving frame will run slower than a clock in a stationary frame. 5 CLO3
- (b) The half-life of a particular particle, as measured in the laboratory comes out to be 4×10^{-8} sec, when its speed is 0.8 c and 3×10^{-8} sec, when its speed is 0.6 c. Explain why this happened. 5 CLO3
- 6 (a) "The length of an object is measured to be shorter when it is moving than when it is at rest." Verify the statement. 5 CLO3
- (b) The length of a spaceship is measured to be exactly half its actual length. Calculate the speed of the spaceship. 5 CLO3

$$F = m \frac{dv}{dt}$$
$$\Rightarrow m \frac{dv}{dt} = F$$

$$e = N \frac{d\phi}{dt}$$

$$L = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$